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SC4003/CE/CZ4046: Intelligent Agents
Assignment 2 Report

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Three-Player Repeated Prisoner's Dilemma

1. Introduction

This report describes the design and evaluation of the **Varghese_Ivan_Player** agent, a strategy for the three-player repeated Prisoner's Dilemma tournament. The strategy is built around the principle of *adaptive graduated retaliation*: start cooperatively, build a statistical model of each opponent's behaviour, and respond proportionately — without over-punishing noisy or temporarily defecting players.

2. Payoff Structure

The three-player payoff for player 1 satisfies the ordering:

$$U(DCC) > U(CCC) > U(DDC) > U(CDC) > U(DDD) > U(CDD)$$

Outcome	Player 1 payoff	Interpretation
DCC	8	Best: player 1 defects, both others cooperate
CCC	6	All cooperate — strong mutual benefit
DDC	5	Two defect, one cooperates
CDC	3	Player 1 cooperates, one opp defects, one coops
DDD	2	All defect — Nash equilibrium
CDD	0	Worst: player 1 cooperates, both others defect

Key observation: being the sole cooperator while both opponents defect yields 0 points (CDD), the absolute worst outcome. Being the sole defector against two cooperators yields 8 points. This asymmetry means a strategy must quickly detect chronic defectors and switch to defection to avoid exploitation.

3. Strategy Design

3.1 Overview

The strategy has five interlocking components:

- **Warm start:** cooperate unconditionally for the first 2 rounds. This seeds goodwill with cooperative opponents and avoids punishing the first (random-looking) move of T4TPlayer or TolerantPlayer.

- **Per-opponent defection rate:** after each round the cumulative fraction of defecting moves is computed independently for each opponent. Using the full history makes the estimate increasingly accurate as the match progresses.
- **Graduated threshold (0.60):** an opponent is classified as a defector only when their defection rate exceeds 60%. This tolerates noisy players (e.g. RandomPlayer, ~50% defection rate) and forgives occasional lapses without triggering an irreversible retaliation spiral.
- **Last-round trigger:** even when an opponent is classified as a defector, we only defect if that opponent also defected in the immediately preceding round. This reactive element prevents sustained punishment of a reformed defector and mirrors the successful tit-for-tat intuition.
- **Periodic forgiveness:** every 15 rounds the player cooperates unconditionally. This gives chronic defectors a chance to cooperate, breaks mutual-defection locks, and recovers value in matches that have temporarily soured.

3.2 Decision Logic (pseudocode)

```

if n < 2: return COOPERATE
if n mod 15 == 0: return COOPERATE # forgiveness
defRate1 = defections(oppHistory1) / n
defRate2 = defections(oppHistory2) / n
opp1Bad = (defRate1 > 0.60) AND oppHistory1[n-1]==DEFECTION
opp2Bad = (defRate2 > 0.60) AND oppHistory2[n-1]==DEFECTION
if opp1Bad OR opp2Bad: return DEFECTION
if defRate1 > 0.60 AND defRate2 > 0.60: return DEFECTION
return COOPERATE

```

3.3 Parameter Justification

The defection threshold of 0.60 was chosen empirically. A value of 0.50 would misclassify RandomPlayer (whose expected rate is 0.50) and lead to unnecessary mutual defection against a player that is roughly neutral. A value above 0.70 would delay retaliation against NastyPlayer too long, conceding many 0-point rounds early in a match. 0.60 sits cleanly between the RandomPlayer baseline and a persistently defecting strategy.

The forgiveness interval of 15 rounds means roughly 6–7 forgiveness events per 100-round match — enough to probe a defector's willingness to reform without giving up too much score when they remain hostile.

4. Expected Performance Against Baseline Strategies

The table below summarises the expected dynamic against each of the six baseline players included in the tournament code.

Opponent	Expected dynamic	Approx. avg. payoff
NicePlayer (always C)	Both cooperate throughout. Mutual CCC payoff.	~6.0
NastyPlayer (always D)	We cooperate 2 rounds, then defect after rate exceeds 0.60. Mostly mutual DD.	~4.6
RandomPlayer (~50% D)	Rate stays near 0.50, under threshold. We cooperate most rounds; they defect ~half.	~3.5–4.0
TolerantPlayer	TolerantPlayer cooperates unless majority defect. We cooperate → mutual CCC.	~5.5–6.0
FreakyPlayer(C)	Decided to cooperate: mutual CCC the whole match.	~6.0
FreakyPlayer(D)	Decided to defect: we retaliate after warmup. Mostly mutual DD.	~4.6
T4TPlayer	T4T mirrors us. We cooperate → it cooperates back. Mutual CCC (mostly).	~5.5–6.0

Against the full baseline field the strategy is expected to place near the top. It matches or beats NicePlayer (which is exploited by NastyPlayer), avoids the zero-payoff disaster of pure cooperation, and sustains near-CCC payoffs against all cooperative or mixed players.

5. Robustness and Limitations

Strengths. The strategy is robust to noisy opponents (RandomPlayer, T4TPlayer) because of the tolerance threshold. It recovers from mutual-defection traps through periodic forgiveness. It handles the three-player extension naturally by tracking each opponent independently.

Limitations. Against an opponent with a very gradual increase in defection rate that stays just below 0.60, the strategy may be slow to retaliate. Against highly sophisticated adaptive strategies designed specifically to exploit the forgiveness windows, the 15-round period could be tuned. These are minor concerns in a one-shot tournament where the opponent distribution is unknown.

6. Conclusion

This agent implements a principled, adaptive cooperation strategy grounded in reputation tracking and graduated retaliation. By combining a tolerance threshold, a reactive last-round trigger, and periodic forgiveness, it achieves high payoffs against cooperative and mixed opponents while avoiding exploitation by persistent defectors. The design philosophy — cooperate by default, retaliates proportionately, forgive occasionally — is well-supported by the theoretical literature on iterated dilemmas and is expected to perform competitively in the tournament.

